Dead-Direction Signatures and Arc Spaces: a Correspondence and its Limits

Setup

$K : \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ is the population loss (KL divergence), with $K(w^*) = 0$, and $W_0 = \{K = 0\}$ is the set of true parameters. λ denotes the RLCT (learning coefficient). For an arc γ, $\mathrm{ord}_\gamma(K)$ is the order of vanishing of $t \mapsto K(\gamma(t))$, and $\mathrm{Cont}^{\geq p}(K) = \{\gamma : \mathrm{ord}_\gamma(K) \geq p\}$, viewed in the jet space $J_{p-1}(\mathbb{R}^d)$ of dimension dp.

The correspondence

Definition 1 (Shirodkar, arXiv:2606.05957)**.** *A unit direction u is a* dead direction of KL order k *at w^* if $K(w^* + tu) = c\,t^{2k} + O(t^{2k+1})$ with $c > 0$.*

Proposition 1. *u is a dead direction of KL order k if and only if the linear arc $\gamma(t) = w^* + tu$ lies in $\mathrm{Cont}^{\geq 2k}(K) \setminus \mathrm{Cont}^{\geq 2k+1}(K)$.*

Proof. Both say $\mathrm{ord}_\gamma(K) = 2k$. □

So dead directions are *linear* arcs sorted by contact order. Directions inside W_0 ($K(w^*+tu) \equiv 0$) are not dead directions; they are arcs of infinite order.

The two formulas for λ

Theorem 1 (DDS selection rule; Shirodkar, Thm. 3)**.** *Under that theorem's hypotheses, a dead direction of KL order k contributes $1/(2k)$ to λ, and contributions of independent directions add.*

Theorem 2 (Mustață; Ein–Lazarsfeld–Mustață, arXiv:math/0102201, math/0303268)**.** *For an ideal $\mathfrak{a}$ (or a function), the complex log canonical threshold is*

$$\mathrm{lct}(\mathfrak{a}) = \min_{p \geq 1} \frac{\mathrm{codim}_{J_{p-1}} \mathrm{Cont}^{\geq p}(\mathfrak{a})}{p}.$$

Two conversions connect this to λ:

- If $K \asymp f_1^2 + \cdots + f_r^2$ near w^*, then $\lambda = \mathrm{RLCT}_{\mathbb{R}}(\mathfrak{a})/2$ for the fiber ideal $\mathfrak{a} = (f_1, \ldots, f_r)$ (Lin, arXiv:1003.5338, Thm. 1.2).
- $\mathrm{RLCT}_{\mathbb{R}}(\mathfrak{a}) = \mathrm{lct}(\mathfrak{a})$ when a minimizing divisor has enough real points. It can fail: for $K = (x^2 + y^2 + z^2)^2$ the ideal $(x^2 + y^2 + z^2)$ has lct $= 1$ but $\lambda = 3/4$. Counting *real* arcs avoids this.

How they relate

DDS is the restriction of the contact formula to linear arcs. The two agree when that restriction loses nothing, as in the smooth-fibre setting of Thm. 3. The arc formula also supplies what Thm. 3 takes as given:

- **Branch selection**: the minimum over contact-locus components;
- **Multiplicity**: the number of minimizing components;
- **Non-linear arcs**: a purely linear reading cannot tell a direction tangent to a curved W_0 from a dead one. For $K = (b - a^2)^2$, K has order 4 along $u = (1, 0)$, so a linear count gives $1/2 + 1/4$. But the arc (t, t^2) stays in W_0, so u is tangent, and $\lambda = 1/2$. (Shirodkar's arXiv:2607.00603 classifies such directions separately.)

Where DDS breaks: crossing points. In reduced-rank regression at the minimum-norm solution, W_0 is a union of components meeting at that point. There $2\lambda - \operatorname{rank} \nabla^2 K = (H - r_0)(N - r_0)$, and this gap is *not* higher-order contact. The contact-locus minimum is reached already at $p = 1$, so λ is half the codimension of the largest component of W_0. The gap is the excess of the tangent space over that component (see `arc/README.md`). Higher-order (quartic) contact, where the DDS count *is* right, appears instead in symmetric factorization $WW^\top$.

Checks

1. $K = a^2b^2$, $u = (1, 0)$: $K(tu) \equiv 0$, an arc inside W_0 (infinite order), not a dead direction.
2. $u = (1, 1)/\sqrt{2}$: $K(tu) = t^4/4$, KL order 2, contribution $1/4$.
3. Contact formula for $K = a^2b^2$ at $p = 2$: $\mathrm{Cont}^{\geq 2} = \{a_0 b_0 = 0\} \subset J_1(\mathbb{R}^2) \cong \mathbb{R}^4$ has codimension 1, so the ratio is $1/2$. That equals $\lambda = 1/2$, and also $\mathrm{lct}(a^2b^2) = 1/2$. Equivalently, with fiber ideal (ab): $\mathrm{lct}(ab) = 1$ and $\lambda = 1/2$. (An earlier version of this note computed $2 - 3$ instead of $2 - 3/2$ here and then explained a non-existent discrepancy.)
4. Branch selection: on $\{a = 0\}$ and on $\{b = 0\}$ the codimension is 1 and $\lambda = 1/2$; at the origin both components are present and the minimum is still $1/2$, with multiplicity 2 when the branches have equal codimension.